

Discrete Mathematics — Homework 3: Inference and First Order Logic (FOL)

Solutions (prepared 2026-02-21)

1. Validity

Question being asked:

Consider the inference rule: $(p \rightarrow q), q \implies p$, where p = "I am cold" and q = "I forgot my jacket". Determine whether it is valid, explain briefly, and verify via a truth table whether $((p \rightarrow q) \wedge q) \rightarrow p$ is a tautology.

Answer:

This inference rule is not valid. It has the invalid form affirming the consequent.

Counterexample: let p be false and q be true. Then $(p \rightarrow q)$ is true and q is true, but p is false; so the premises can be true while the conclusion is false.

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\[
\begin{array}{c|c|c|c|c}
p & q & (p \rightarrow q) & q & ((p \rightarrow q) \wedge q) \rightarrow p \\
\hline
T & T & T & T & T \\
T & F & F & F & T \\
F & T & T & T & F \\
F & F & T & F & T
\end{array}
\]
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Since the final column is false when $p = F$ and $q = T$, the formula is not a tautology.

2. Inference

Question being asked:

Given premises: 1) $\text{rainy} \rightarrow (\text{cold} \vee \text{windy})$, 2) $\neg \text{cold}$, 3) $\text{windy} \rightarrow \text{uncomfortable}$, 4) $\neg \text{uncomfortable}$, 5) $\text{stormy} \rightarrow \text{rainy}$. Prove $\neg \text{stormy}$, citing each rule/law.

Answer (derivation):

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\[
\begin{array}{r l l}
1. & \text{rainy} \rightarrow (\text{cold} \vee \text{windy}) & \text{\textit{Premise}} \\
2. & \neg \text{cold} & \text{\textit{Premise}} \\
3. & \text{windy} \rightarrow \text{uncomfortable} & \text{\textit{Premise}} \\
4. & \neg \text{uncomfortable} & \text{\textit{Premise}} \\
5. & \text{stormy} \rightarrow \text{rainy} & \text{\textit{Premise}} \\
6. & \text{\textit{Assume } stormy} & \text{\textit{Assumption for contradiction}} \\
7. & \text{rainy} & \text{5, 6 \textit{Modus Ponens}} \\
8. & \text{cold} \vee \text{windy} & \text{1, 7 \textit{Modus Ponens}} \\
9. & \text{windy} & \text{8, 2 \textit{Disjunctive Syllogism}} \\
10. & \text{uncomfortable} & \text{3, 9 \textit{Modus Ponens}} \\
11. & \bot & \text{10, 4 \textit{Contradiction}} \\
12. & \neg \text{stormy} & \text{6 \textit{Negation Introduction}}
\end{array}
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\end{array}
\]
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3. FOL Propositions (Books domain)

Question being asked:

Let the domain be $B = \text{"Books"}$. Translate the given English sentences into FOL using only the predicates $\text{novel}(x)$, $\text{bestselling}(x)$, $\text{engaging}(x)$, $\text{wellwritten}(x)$, $\text{textbook}(x)$, $\text{difficult}(x)$, $\text{biography}(x)$, $\text{informative}(x)$.

Answer:

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\begin{aligned}
(1)\;&\forall x \big((\text{novel}(x) \wedge \text{bestselling}(x)) \rightarrow (\text{engaging}(x) \wedge \text{wellwritten}(x))\big) \\
(2)\;&\exists x \big(\text{textbook}(x) \wedge \neg \text{difficult}(x)\big). \quad [2pt] \\
(3)\;&\big(\forall x \big(\text{biography}(x) \rightarrow \text{informative}(x)\big)\big) \wedge \\
&\quad \big(\exists x \big(\text{novel}(x) \wedge \text{bestselling}(x)\big)\big) \\
\end{aligned}
\]
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4. FOL Propositions (People and Movies domains)

Question being asked:

Let the domain be $P = \text{"People"}$ and $M = \text{"Movies"}$. Translate (a)–(e) into FOL using predicates $\text{comedy}(y)$, $\text{drama}(y)$, $\text{enjoys}(x,y)$, $\text{directed}(x,y)$, $\text{acclaimed}(y)$.

Answer:

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\begin{aligned}
(a)\;&\forall x \exists y \big(\text{comedy}(y) \wedge \text{enjoys}(x,y)\big). \quad [2pt] \\
(b)\;&\exists y \big(\text{drama}(y) \wedge \forall x \text{enjoys}(x,y)\big). \quad [2pt] \\
(c)\;&\forall x \exists y \big(\text{comedy}(y) \wedge \neg \text{enjoys}(x,y)\big). \quad [2pt] \\
(d)\;&\forall x \forall y \big(\text{directed}(x,y) \rightarrow \text{enjoys}(x,y)\big). \quad [2pt] \\
(e)\;&\forall y \big(\text{acclaimed}(y) \rightarrow \exists x \text{directed}(x,y)\big). \\
\end{aligned}
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5. Negation of FOL Propositions

Question being asked:

For each formula, write an equivalent formula by pushing negations inward until each $\neg$ appears directly before a predicate.

Answer:

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\begin{aligned}
(a)\;&\neg(\forall x, (P(x) \rightarrow Q(x))) \\
&\equiv \exists x, \neg(P(x) \rightarrow Q(x)) \\
&\equiv \exists x, (P(x) \wedge \neg Q(x)). \quad [6pt] \\
(b)\;&\neg(\exists x, (R(x) \wedge \forall y, S(x,y))) \\
&\equiv \forall x, \neg(R(x) \wedge \forall y, S(x,y)) \\
&\equiv \forall x, (\neg R(x) \vee \neg \forall y, S(x,y)) \\
&\equiv \forall x, (\neg R(x) \vee \exists y, \neg S(x,y)). \quad [6pt]
\end{aligned}
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(c)\;& \not(\forall x\,(\exists y\,P(x,y)\to \forall z\,Q(z)))\
&\equiv \exists x\,(\exists y\,P(x,y)\land \exists z\,\not Q(z)).
\end{aligned}
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